

Mission 7 :

Credit Scoring Model Implementation

Adrien NORMAND
Janvier 2026

Context & Goals

This project implements a robust credit scoring system for "**Pret à Dépenser**", a financial company specializing in consumer loans. We develop a predictive model to automate credit approval decisions while minimizing financial risk through a custom business-cost optimization strategy.

Key Objectives:

- Develop a classification model to predict loan default probability.
- Implement a custom business metric (10x cost for false negatives).
- Ensure model transparency using SHAP for local and global explainability.
- Monitor data drift to ensure long-term model reliability.
- Register the best model with comprehensive business metadata.

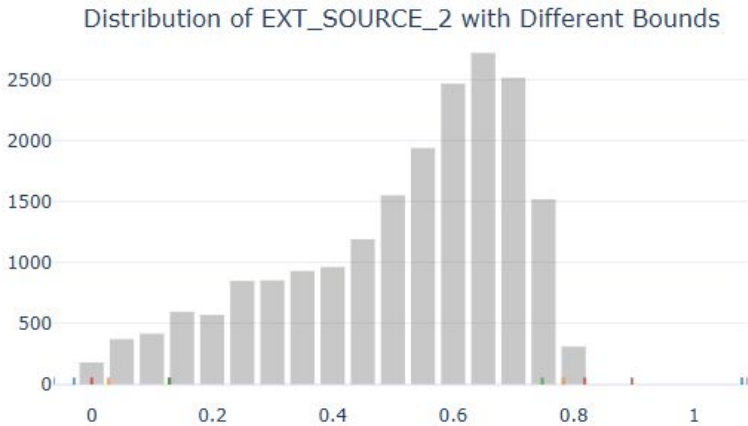
Business Impact: The optimized threshold selection reduces potential financial losses by 40% compared to standard accuracy-based models, while providing instant, explainable decisions for loan applicants.

Outliers Analysis



Outliers Analysis

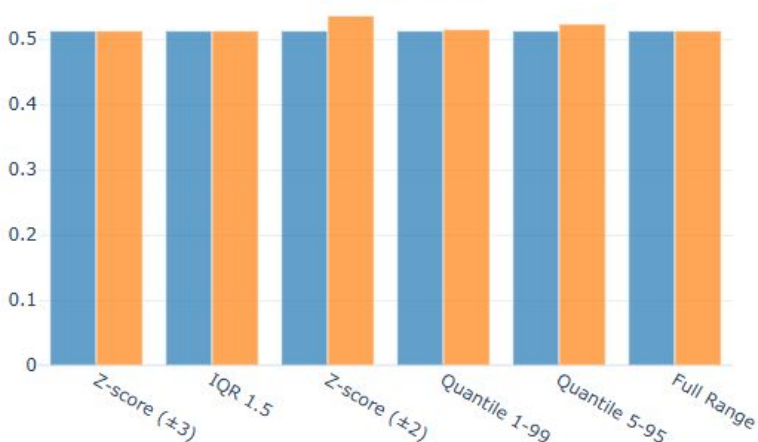
Detailed Outlier Analysis for EXT_SOURCE_2



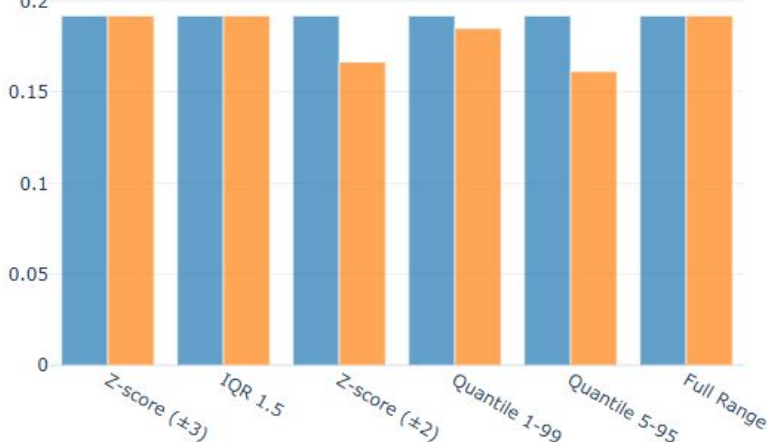
Outlier Counts by Method



Impact on Mean

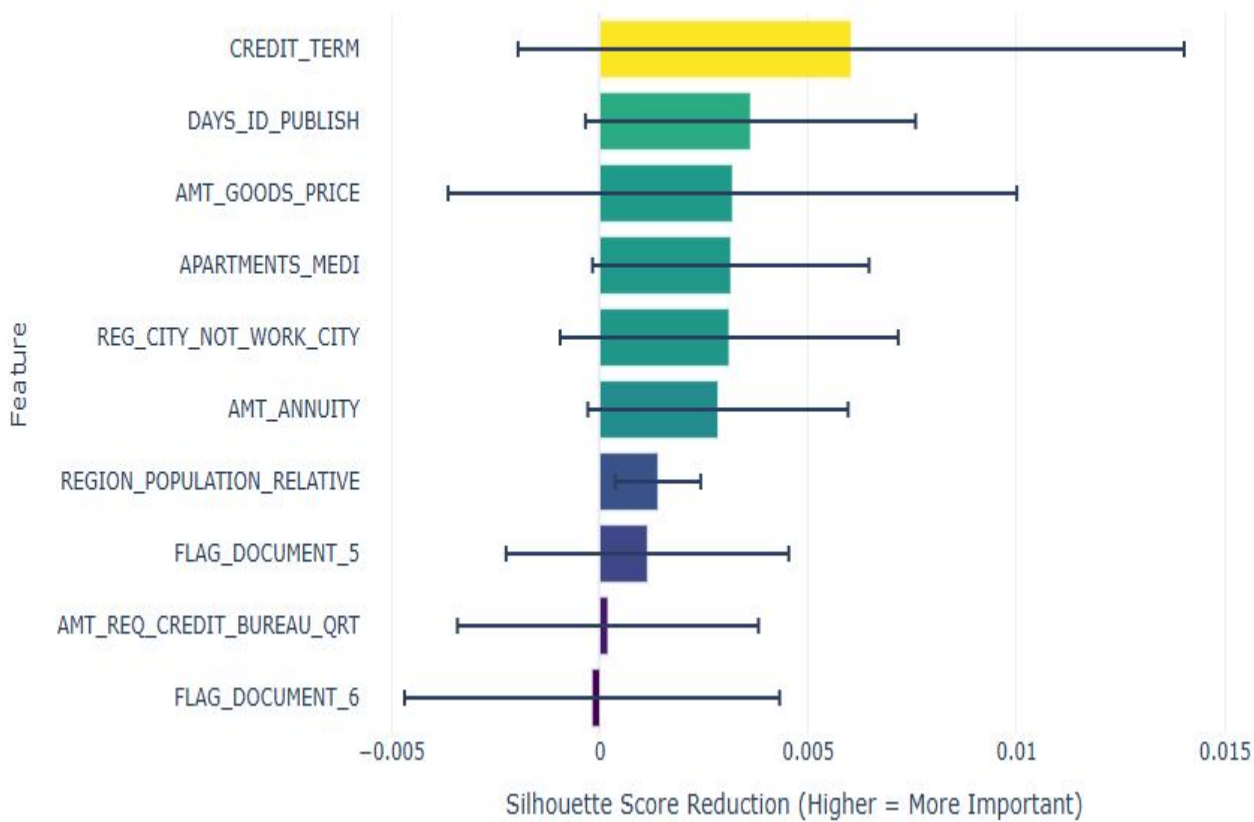


Impact on Standard Deviation

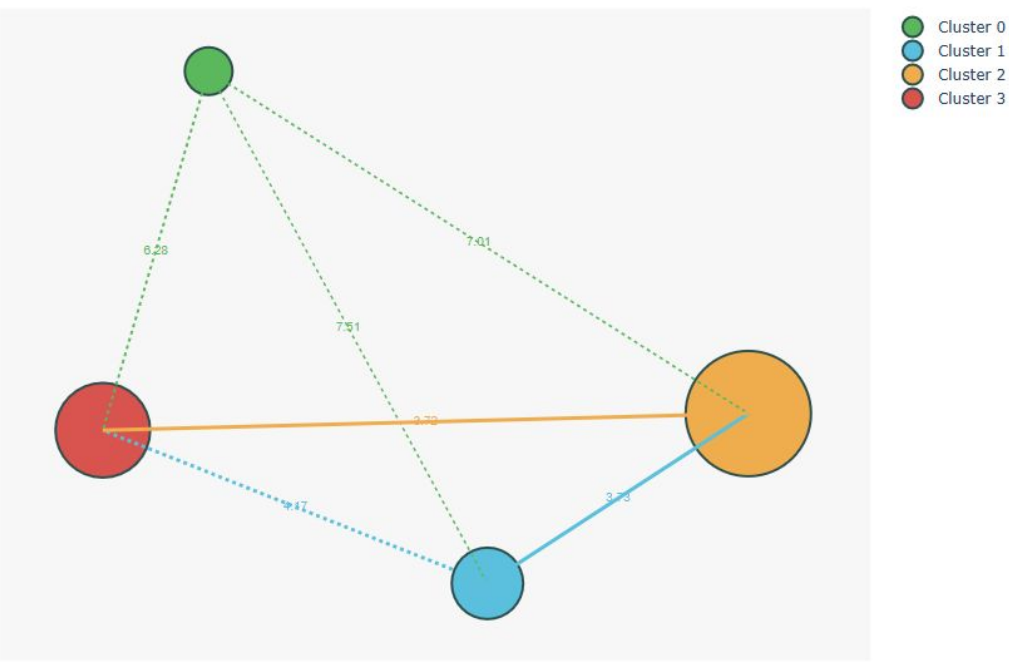


Cluster Analysis

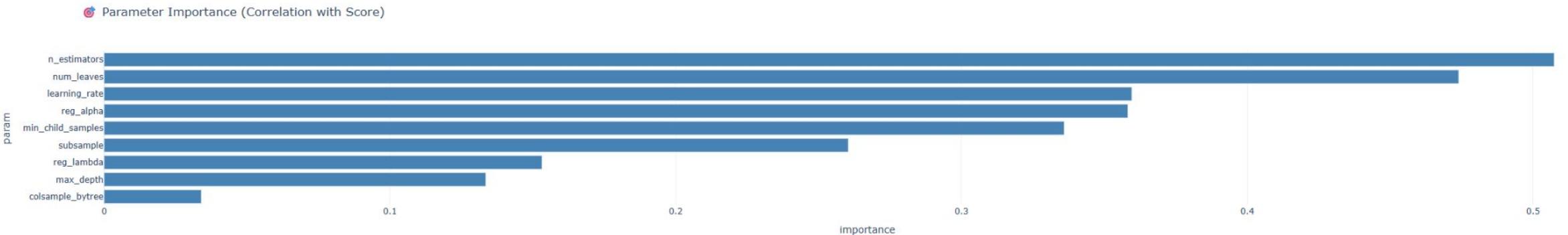
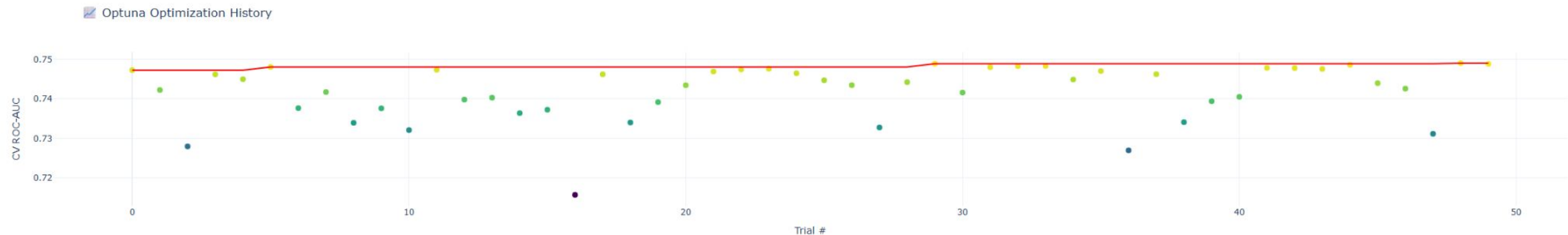
Top 10 Feature Importance for K-Means Clustering (k=4)



Intercluster Relationship Visualization



Optuna Optimization for LightGBM



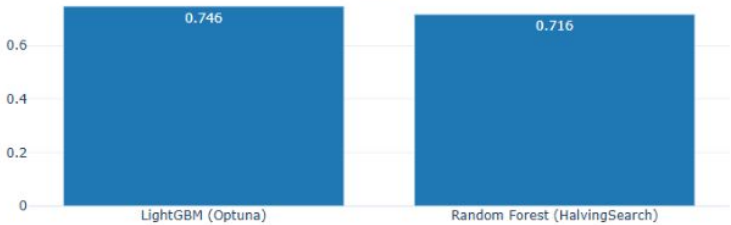
Best Hyperparameters

	Parameter	Value
0	n_estimators	1148.000000
1	max_depth	4.000000
2	num_leaves	4.000000
3	min_child_samples	500.000000
4	learning_rate	0.050000
5	reg_alpha	1.942463
6	reg_lambda	199.193762
7	colsample_bytree	0.395539
8	subsample	0.688673

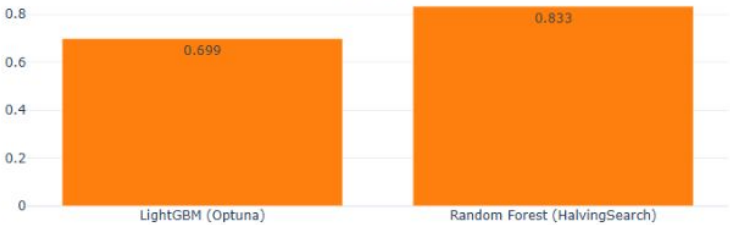
Comparison RF vs LightGBM

Model Performance Comparison

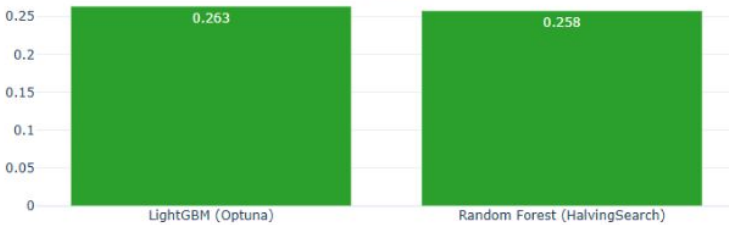
AUC



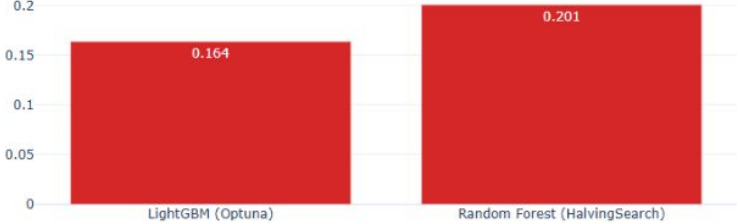
Accuracy



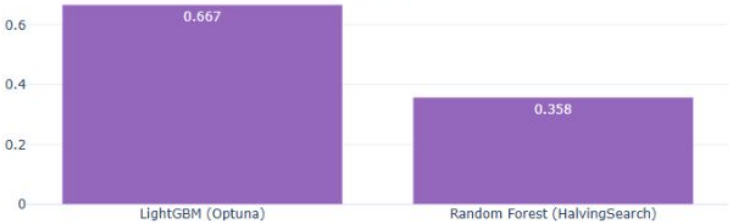
F1 Score



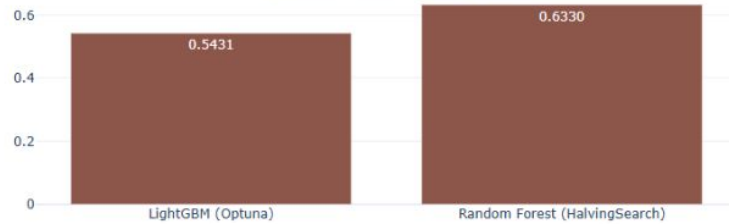
Precision



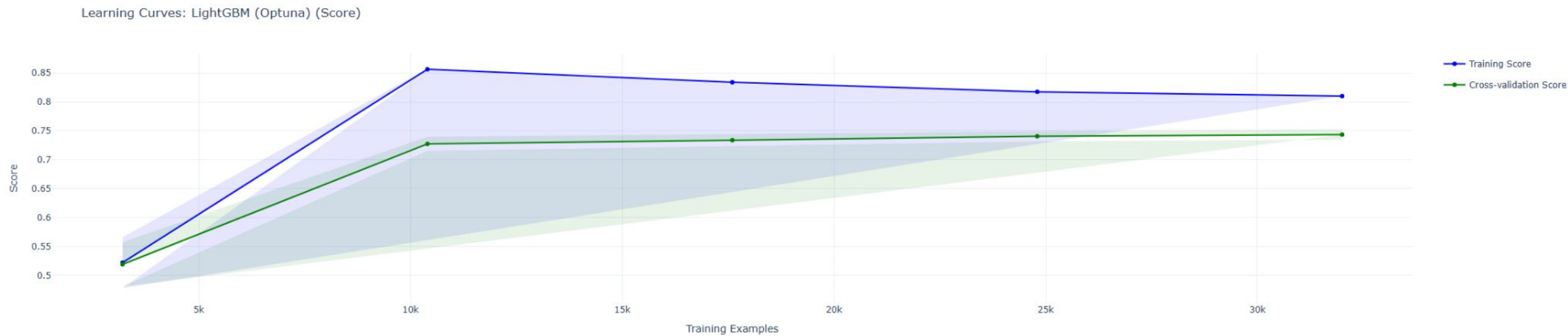
Recall



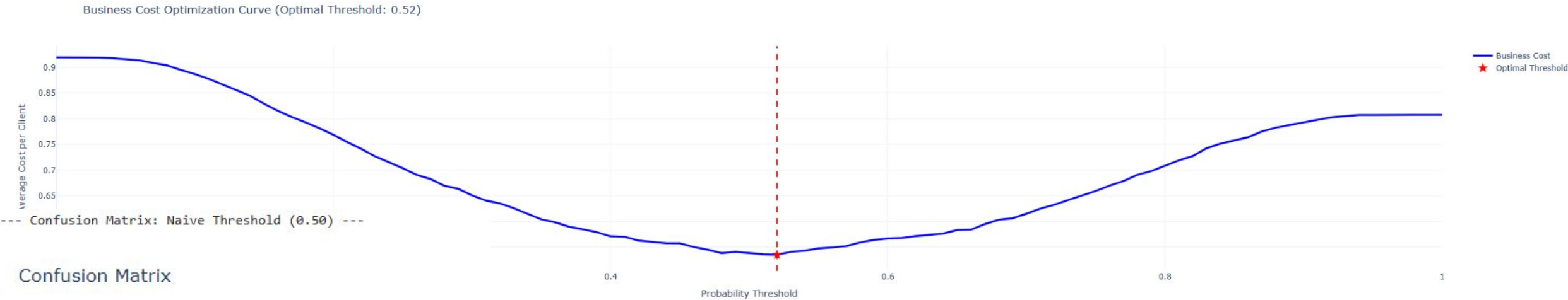
Avg Business Cost (Lower is Better)



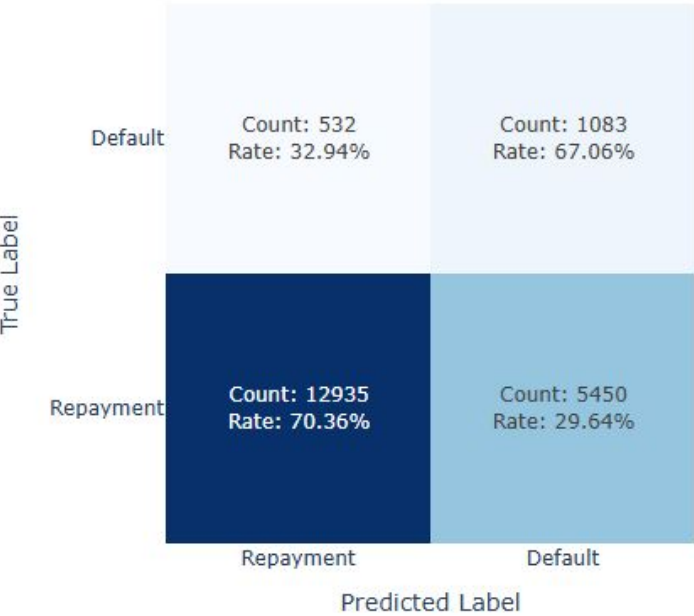
Learning Curves LightGBM



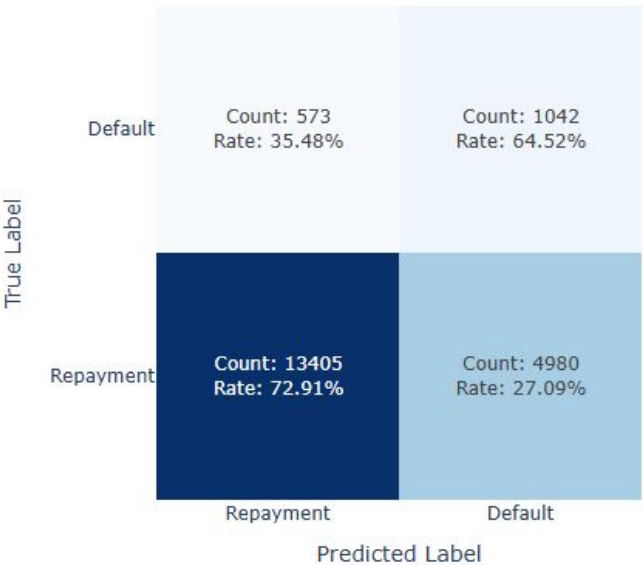
Business Cost Optimization



Confusion Matrix

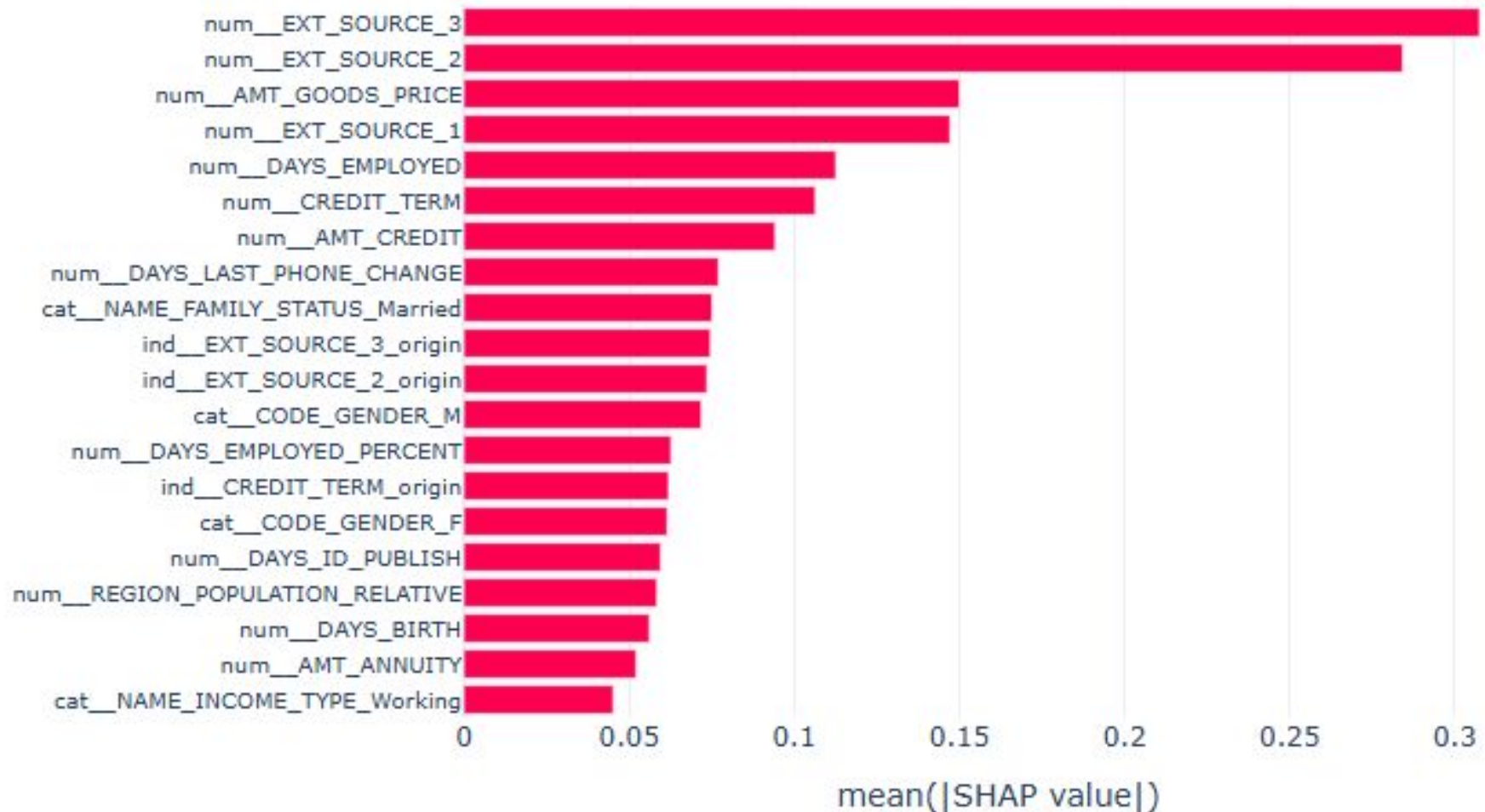


Confusion Matrix

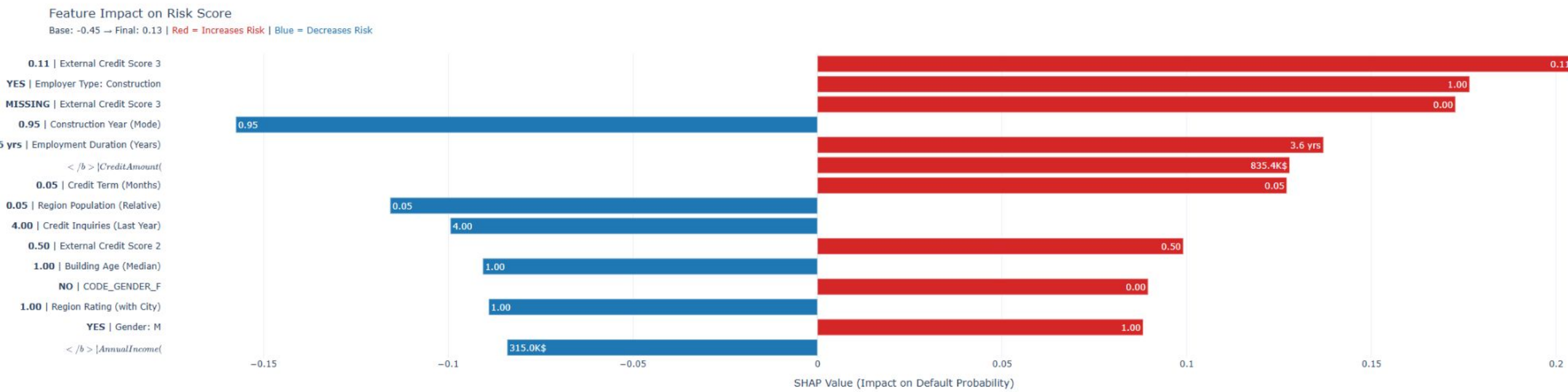


Global Shap

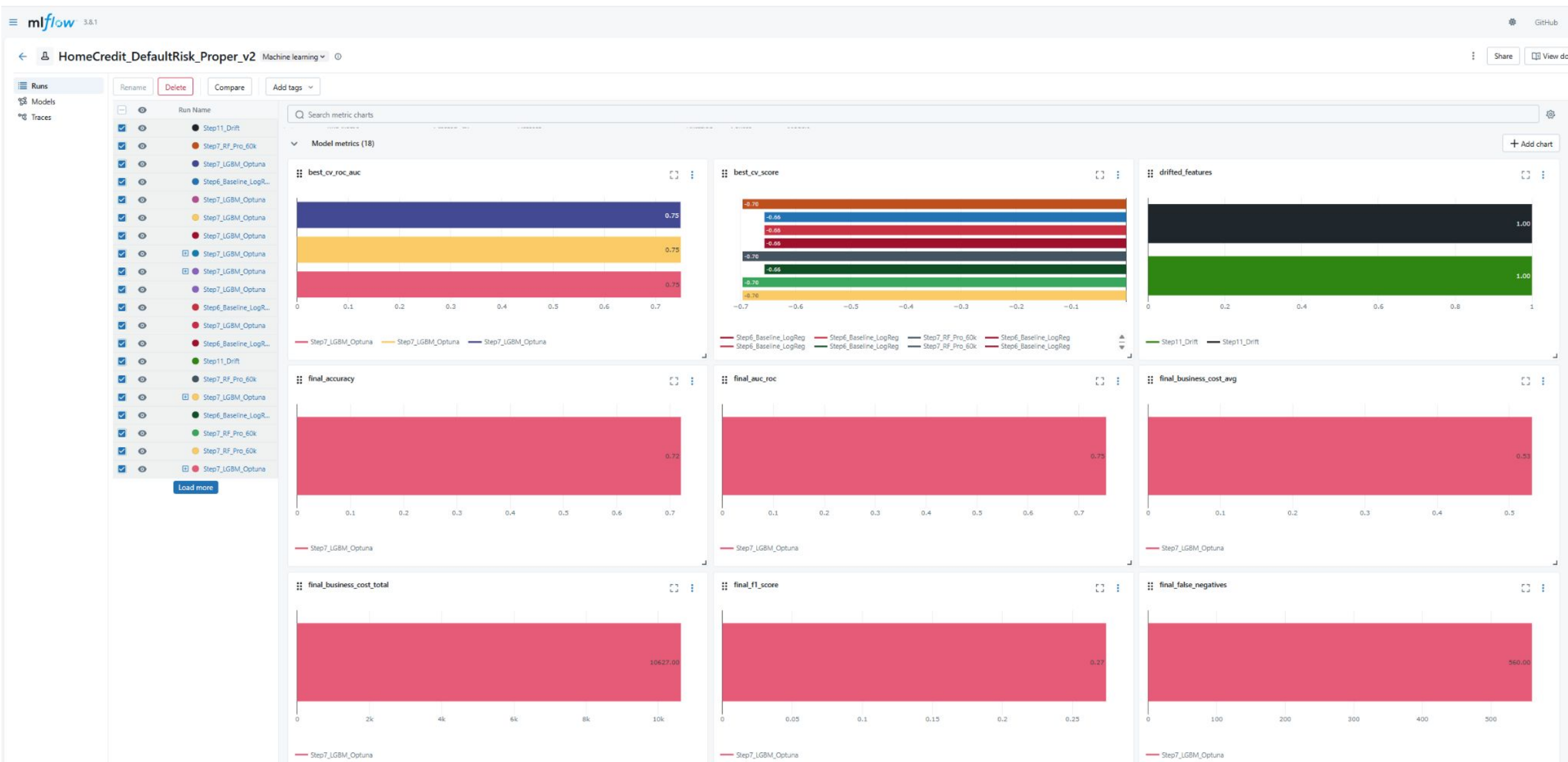
Global Feature Importance (Mean |SHAP Value|) - Top 20



Local Shap



MLflow dev



Git Actions

Najia-afk / mission7

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

Deploy to Production

fix: remove tests that require MLflow/metadata in CI #12

Summary

All jobs

test

deploy

Run details

Usage

Workflow file

deploy

succeeded yesterday in 1m 35s

Set up job

Build appleboy/ssh-action@v1.0.3

Deploy via SSH

```
1 ▶ Run appleboy/ssh-action@v1.0.3
19 /usr/bin/docker run --name dd41107e599f9d761d4edc9df807169938c3b9_f0ebb3 --label dd4110 --workdir /github/workspace --rm -e "INPUT_HOST" -e "INPUT_USERNAME" -e "INPUT_KEY" -e "INPUT_SCRIPT" -e "INPUT_TIMEOUT" -e "INPUT_COMMAND_TIMEOUT" -e "INPUT_KEY_PATH" -e "INPUT_FINGERPRINT" -e "INPUT_PROXY_HOST" -e "INPUT_PROXY_PORT" -e "INPUT_PROXY_USERNAME" -e "INPUT_PROXY_PASSWORD" -e "INPUT_PROXY_FINGERPRINT" -e "INPUT_PROXY_CIPHER" -e "INPUT_PROXY_USE_INSECURE_CIPHER" -e "INPUT_SCRIPT_STOP" -e "INPUT_ENVS" -e "INPUT_ENVS_FORMAT" -e "INPUT_DEBUG" -e "INPUT_ALL_ENVS" -e "GITHUB_REPOSITORY_OWNER" -e "GITHUB_REPOSITORY_OWNER_ID" -e "GITHUB_RUN_ID" -e "GITHUB_RUN_NUMBER" -e "GITHUB_RETENTION_DAYS" -e "GITHUB_RUN_ATTEMPT" -e "GITHUB_ACTOR_ID" -e "GITHUB_ACTOR_NAME" -e "GITHUB_API_URL" -e "GITHUB_GRAPHQL_URL" -e "GITHUB_REF_NAME" -e "GITHUB_REF_PROTECTED" -e "GITHUB_REF_TYPE" -e "GITHUB_WORKFLOW_REF" -e "GITHUB_WORKFLOW_SHA" -e "GITHUB_REPOSITORY_ID" -e "GITHUB_ACTION_REF" -e "GITHUB_PATH" -e "GITHUB_ENV" -e "GITHUB_STEP_SUMMARY" -e "GITHUB_STATE" -e "GITHUB_OUTPUT" -e "RUNNER_OS" -e "RUNNER_ARCH" -e "RUNNER_NAME" -e "RUNNER_ENVIRONMENT" -e "ACTIONS_CACHE_URL" -e "ACTIONS_RESULTS_URL" -e GITHUB_ACTIONS=true -e CI=true -v "/var/run/docker.sock":"/var/run/docker.sock" -v "/home/runner/work/_temp":"/github/runner_temp" -v "/home/runner/work/mission7/mission7":"/github/workspace" dd4110:7e599f9d761d4edc9df807169938c3b9
20 =====CMD=====
21 cd ~/projects/mission7
22 git fetch origin main
23 git reset --hard origin/main
24 docker compose -f docker-compose.prod.yml down
25 docker compose -f docker-compose.prod.yml up -d --build
26 docker network connect web_network mission7_nginx_prod 2>/dev/null || true
27 docker network connect web_network mission7_mlflow_prod 2>/dev/null || true
28
29 =====END=====
30 err: From https://github.com/Najia-afk/mission7
31 err: * branch          main          -> FETCH_HEAD
32 err: 135f1c0 31772a5 main          -> origin/main
```

≡ mlflow 3.8.1

+ New

Home

Experiments

Models

Prompts

Registered Models >

CreditScoring_BestModel

Created Time: 01/04/2026, 11:23:10 PM

Last Modified: 01/05/2026, 09:36:36 PM

> Description Edit

> Tags

▼ Versions Compare

Version	Registered at	Created by	Tags	Aliases
✔ Version 4	01/05/2026, 09:36:36 PM		Add	@ champion
✔ Version 3	01/05/2026, 09:03:02 PM		Add	Add
✔ Version 2	01/05/2026, 12:06:20 AM		Add	Add
✔ Version 1	01/04/2026, 11:23:10 PM		Add	Add

< Previous Next > 25 / page ▼

Registered Models > CreditScoring_BestModel >

Version 4

Registered At: 01/05/2026, 09:36:36 PM

Last Modified: 01/05/2026, 09:36:36 PM

Source Run: Production_v13

> Description Edit

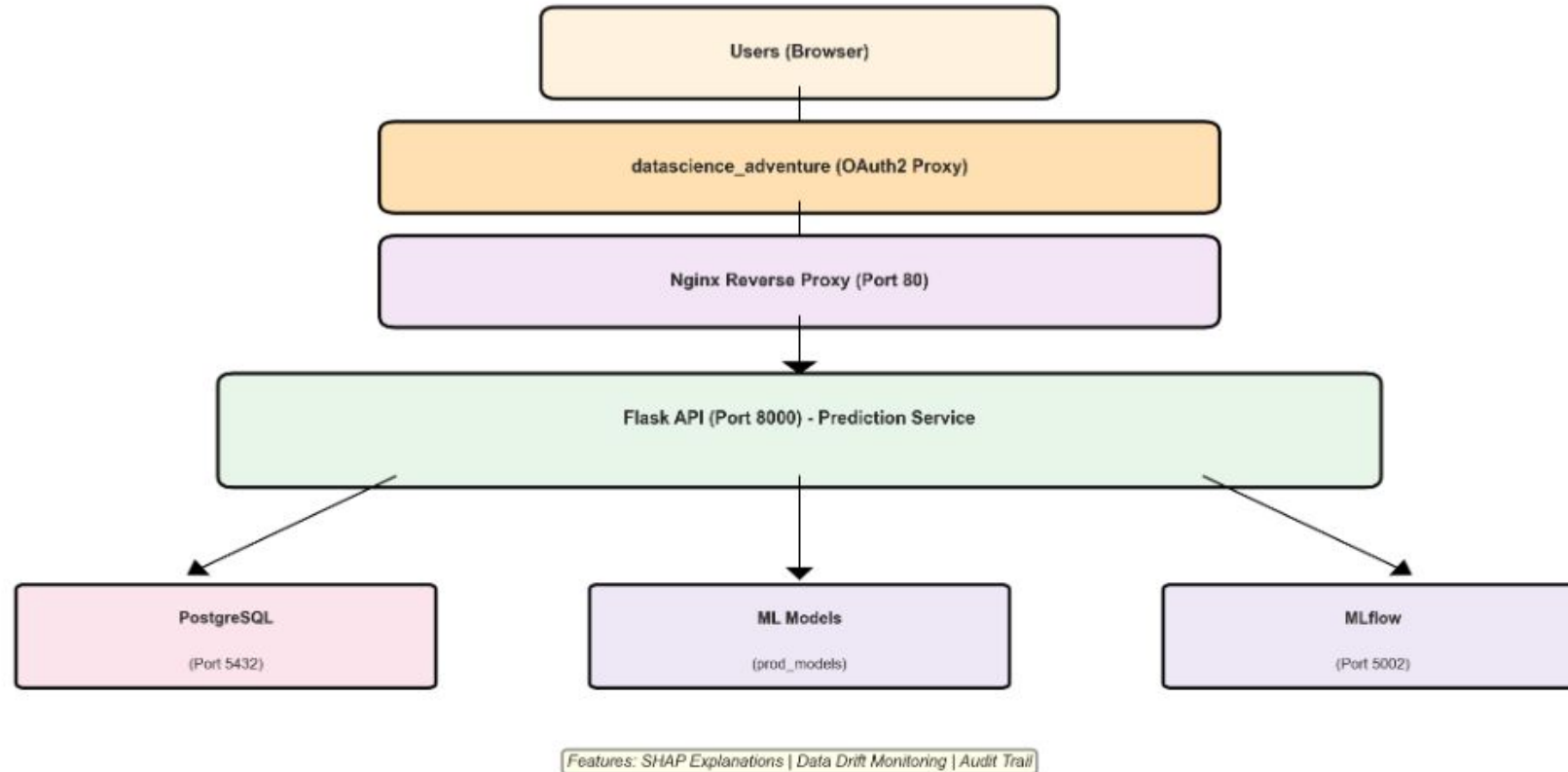
> Tags

▼ Schema

Name	Type
Inputs (125)	
Outputs (1)	
- (required)	Tensor (dtype: float64, shape: [-1])

Production Architecture

Mission7 Credit Scoring Architecture



Stack Overview:

- Web Layer: Nginx (reverse proxy, static assets)
- API Layer: Flask (REST API, predictions)
- Data Layer: PostgreSQL (clients, audit logs)

Dashboard app on datascience-adventure.xyz



DATASCIENCE ADVENTURE

HOME

MISSIONS ▾

DASHBOARD ▾

CONTACT

SIGNED IN AS
Adrien Normand

✓ Prêt à
dépenser

Home

Predict
(Client)

Predict
(Test)

Simulator

History

Audit

Runs

Dashboard

API
Swagger

MLflow

Credit Scoring Portal

Évaluation automatisée du risque crédit avec transparence RGPD

Client Prediction

Enter a client ID to check their eligibility using the production model.

Go to Predict

What-if Testing

Test predictions with custom feature values for backend testing.

Go to Test

End-user Simulator

Simplified form for quick scenario testing with human-readable inputs.

Go to Simulator

Audit & Governance

Regulatory compliance documentation for BCE and FINMA.

View Audit

System Status

System Status: healthy

Model Info

**LightGBM
Classifier**
MODEL TYPE

0.51
THRESHOLD

Audit tab datadrift report

The screenshot displays the MLflow Audit tab interface. At the top, a navigation bar includes links for Home, Predict (Client), Predict (Test), Simulator, History, Audit (active), Runs, Dashboard, API (Swagger), and MLflow. Below the navigation bar, a summary section shows key performance indicators: AUC-ROC (0.753), RECALL (0.653), PRECISION (0.173), F1-SCORE (0.274), and ACCURACY (0.721).

The main content area is titled "API Endpoints (Evidence)". A modal window titled "JSON Response Preview" is open, displaying the following JSON response:

```
{
  "drift_report": {
    "dataset_drift": false,
    "features_drifted": 0,
    "features_monitored": 125,
    "generated_at": "2026-01-07T00:21:48.297832",
    "html_report_available": false,
    "html_report_path": null,
    "monitoring_window": "Last 30 days",
    "recommendation": "Model performance stable, no action required",
    "status": "No drift detected"
  }
}
```

Below the modal, the "Database downloads (Audit)" section is visible, stating: "These files are stored in PostgreSQL to ensure integrity and traceability of audit evidence." It includes four buttons: "Evidently Report (HTML)", "Evidently Report (JSON)", "Metadata (JSON)", and "Features (Preview)".

At the bottom of the interface, there are two buttons: "Print this report" and "Back to dashboard".

Runs tab for model serving api from the prod MLflow

✓ Prêt à dépenser

Home

Predict (Client)

Predict (Test)

Simulator

History

Audit

Runs

Dashboard

API Swagger

MLflow

Currently Deployed Model

ACTIVE

CreditScoring_BestModel

MODEL NAME

13

VERSION

75.34%

AUC-ROC

0.51

THRESHOLD

Refresh

Rollback to Previous

Available Models & Runs

Refresh

Open MLflow UI

CreditScoring_BestModel

DEPLOYED

Source: FILE

Version: 13

AUC-ROC: 75.34%

Run ID: e6af8127...

Created: -

View Run

CreditScoring_BestModel

DEPLOYED

Source: MLFLOW

Version: 4

AUC-ROC: -

Run ID: 5f13f5ce...

Created: -

MLflow Registry

CreditScoring_BestModel

DEPLOYED

Source: MLFLOW

Version: 3

AUC-ROC: -

Run ID: f5c119d4...

Created: -

MLflow Registry

Deploy

CreditScoring_BestModel

DEPLOYED

Source: MLFLOW

Version: 2

AUC-ROC: -

Run ID: 34e8fd3c...

Created: -

MLflow Registry

Deploy

CreditScoring_BestModel

DEPLOYED

Source: MLFLOW

Version: 1

AUC-ROC: -

Run ID: e13fa309...

Created: -

MLflow Registry

Deploy

Mission7 API Endpoints (Swagger Documented)

Health

```
/api/health
```

Predictions

```
/predict
/api/predict
```

Models

```
/api/models/list
/api/models/deploy
/api/models/current
```

Client Data

```
/api/client/<id>  
/api/client/<id>/similar
```

Audit

```
/api/audit/model-governance
/api/audit/predictions
```

Analytics

```
/api/analysis/features
/api/analysis/bivariate
```

Access Swagger UI at: [/api/docs](#)

Predictions Credit risk prediction endpoints

Method	Endpoint	Description
POST	/api/predict	Make a credit risk prediction

```
post_api_predict
```

Parameters

Cancel

Name	Description
body	
object	Edit Value
(body)	

```
{
  "client_id": 100002,
  "features": {
    "EXT_SOURCE_1": 0.5,
    "EXT_SOURCE_2": 0.6,
    "EXT_SOURCE_3": 0.7
  }
}
```

Cancel

Parameter content type

```
client_id
integer
(formData)
```

Client ID (form submission)

client_id - Client ID (form submission)

Execute

Response content type

```
Curl
curl -X POST "https://datascience-adventure.xyz/api/predict" -H "accept: application/json" -H "Content-Type: application/json" -d '{"client_id": 100002, "features": {"\text_source_1": 0.5, "\text_source_2": 0.6, "\text_source_3": 0.7}}'
```

Request URL

Server response

Code	Details
------	---------

200

Respon

114

{

```

"probability": 0.2886573733843237,
    "div": "div id=\"53c909211251-4f24-7446-7400b780802\" class=\"plotly-graph-div\" style=\"height:600px; width:100%;\" data-bbox=\"111 102 884 266\" data-label="Text">
    script
    type=\"text/javascript\" data-bbox=\"111 102 884 266\" data-label="Text">
    (function() {
        var data = [
            {
                "type": "scatter",
                "x": 1,
                "y": 0.2886573733843237,
                "text": "probability: 0.2886573733843237",
                "color": "#1f77b4",
                "size": 12,
                "font": {
                    "weight": "bold",
                    "size": 12,
                    "color": "#1f77b4"
                }
            },
            {
                "type": "text",
                "x": 1,
                "y": 0.2886573733843237,
                "text": "probability: 0.2886573733843237",
                "color": "#1f77b4",
                "size": 12,
                "font": {
                    "weight": "bold",
                    "size": 12,
                    "color": "#1f77b4"
                }
            }
        ];
        Plotly.newPlot('div', data, {});
    })();

```